

# Internal Dependencies

## References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [Neo4j Python Driver](#)

## Artifacts

List the artifacts this notebook is based on. Different sorting variations help finding artifacts by their features and support larger code bases where the list of all artifacts gets too long.

Only the top 30 entries are shown. The whole table can be found in the following CSV report:

List\_all\_Java\_artifacts

Table 1a - Top 30 artifacts with the highest package count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-messaging-5.0.3.jar                 | 59       | 579   | 7                    | 2                    |
| 1  | axon-common-5.0.3.jar                    | 15       | 156   | 10                   | 0                    |
| 2  | axon-eventsourcing-5.0.3.jar             | 7        | 100   | 3                    | 4                    |
| 3  | axon-modelling-5.0.3.jar                 | 7        | 92    | 2                    | 3                    |
| 4  | axon-spring-boot-autoconfigure-5.0.3.jar | 7        | 75    | 1                    | 7                    |
| 5  | axon-test-5.0.3.jar                      | 6        | 78    | 1                    | 3                    |
| 6  | axon-conversion-5.0.3.jar                | 5        | 35    | 4                    | 1                    |
| 7  | axon-server-connector-5.0.3.jar          | 5        | 72    | 1                    | 4                    |
| 8  | axon-update-5.0.3.jar                    | 5        | 23    | 1                    | 1                    |
| 9  | axon-metrics-micrometer-5.0.3.jar        | 3        | 16    | 0                    | 3                    |
| 10 | axon-tracing-opentelemetry-5.0.3.jar     | 1        | 5     | 0                    | 2                    |

Table 1b - Top 30 artifacts with the highest type count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-messaging-5.0.3.jar                 | 59       | 579   | 7                    | 2                    |
| 1  | axon-common-5.0.3.jar                    | 15       | 156   | 10                   | 0                    |
| 2  | axon-eventsourcing-5.0.3.jar             | 7        | 100   | 3                    | 4                    |
| 3  | axon-modelling-5.0.3.jar                 | 7        | 92    | 2                    | 3                    |
| 4  | axon-test-5.0.3.jar                      | 6        | 78    | 1                    | 3                    |
| 5  | axon-spring-boot-autoconfigure-5.0.3.jar | 7        | 75    | 1                    | 7                    |
| 6  | axon-server-connector-5.0.3.jar          | 5        | 72    | 1                    | 4                    |
| 7  | axon-conversion-5.0.3.jar                | 5        | 35    | 4                    | 1                    |
| 8  | axon-update-5.0.3.jar                    | 5        | 23    | 1                    | 1                    |
| 9  | axon-metrics-micrometer-5.0.3.jar        | 3        | 16    | 0                    | 3                    |
| 10 | axon-tracing-opentelemetry-5.0.3.jar     | 1        | 5     | 0                    | 2                    |

Table 1c - Top 30 artifacts with the highest number of incoming dependencies

The following table lists the top 30 artifacts that are used the most by other artifacts (highest count of incoming dependencies, highest in-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-common-5.0.3.jar                    | 15       | 156   | 10                   | 0                    |
| 1  | axon-messaging-5.0.3.jar                 | 59       | 579   | 7                    | 2                    |
| 2  | axon-conversion-5.0.3.jar                | 5        | 35    | 4                    | 1                    |
| 3  | axon-eventsourcing-5.0.3.jar             | 7        | 100   | 3                    | 4                    |
| 4  | axon-modelling-5.0.3.jar                 | 7        | 92    | 2                    | 3                    |
| 5  | axon-server-connector-5.0.3.jar          | 5        | 72    | 1                    | 4                    |
| 6  | axon-spring-boot-autoconfigure-5.0.3.jar | 7        | 75    | 1                    | 7                    |
| 7  | axon-test-5.0.3.jar                      | 6        | 78    | 1                    | 3                    |
| 8  | axon-update-5.0.3.jar                    | 5        | 23    | 1                    | 1                    |
| 9  | axon-metrics-micrometer-5.0.3.jar        | 3        | 16    | 0                    | 3                    |
| 10 | axon-tracing-opentelemetry-5.0.3.jar     | 1        | 5     | 0                    | 2                    |

## Table 1d - Top 30 artifacts with the highest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the highest number of other artifacts (highest count of outgoing dependencies, highest out-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-spring-boot-autoconfigure-5.0.3.jar | 7        | 75    | 1                    | 7                    |
| 1  | axon-eventsourcing-5.0.3.jar             | 7        | 100   | 3                    | 4                    |
| 2  | axon-server-connector-5.0.3.jar          | 5        | 72    | 1                    | 4                    |
| 3  | axon-metrics-micrometer-5.0.3.jar        | 3        | 16    | 0                    | 3                    |
| 4  | axon-modelling-5.0.3.jar                 | 7        | 92    | 2                    | 3                    |
| 5  | axon-test-5.0.3.jar                      | 6        | 78    | 1                    | 3                    |
| 6  | axon-messaging-5.0.3.jar                 | 59       | 579   | 7                    | 2                    |
| 7  | axon-tracing-opentelemetry-5.0.3.jar     | 1        | 5     | 0                    | 2                    |
| 8  | axon-conversion-5.0.3.jar                | 5        | 35    | 4                    | 1                    |
| 9  | axon-update-5.0.3.jar                    | 5        | 23    | 1                    | 1                    |
| 10 | axon-common-5.0.3.jar                    | 15       | 156   | 10                   | 0                    |

## Table 1e - Top 30 artifacts with the lowest package count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-tracing-opentelemetry-5.0.3.jar     | 1        | 5     | 0                    | 2                    |
| 1  | axon-metrics-micrometer-5.0.3.jar        | 3        | 16    | 0                    | 3                    |
| 2  | axon-conversion-5.0.3.jar                | 5        | 35    | 4                    | 1                    |
| 3  | axon-server-connector-5.0.3.jar          | 5        | 72    | 1                    | 4                    |
| 4  | axon-update-5.0.3.jar                    | 5        | 23    | 1                    | 1                    |
| 5  | axon-test-5.0.3.jar                      | 6        | 78    | 1                    | 3                    |
| 6  | axon-eventsourcing-5.0.3.jar             | 7        | 100   | 3                    | 4                    |
| 7  | axon-modelling-5.0.3.jar                 | 7        | 92    | 2                    | 3                    |
| 8  | axon-spring-boot-autoconfigure-5.0.3.jar | 7        | 75    | 1                    | 7                    |
| 9  | axon-common-5.0.3.jar                    | 15       | 156   | 10                   | 0                    |
| 10 | axon-messaging-5.0.3.jar                 | 59       | 579   | 7                    | 2                    |

## Table 1f - Top 30 artifacts with the lowest type count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-tracing-opentelemetry-5.0.3.jar     | 1        | 5     | 0                    | 2                    |
| 1  | axon-metrics-micrometer-5.0.3.jar        | 3        | 16    | 0                    | 3                    |
| 2  | axon-update-5.0.3.jar                    | 5        | 23    | 1                    | 1                    |
| 3  | axon-conversion-5.0.3.jar                | 5        | 35    | 4                    | 1                    |
| 4  | axon-server-connector-5.0.3.jar          | 5        | 72    | 1                    | 4                    |
| 5  | axon-spring-boot-autoconfigure-5.0.3.jar | 7        | 75    | 1                    | 7                    |
| 6  | axon-test-5.0.3.jar                      | 6        | 78    | 1                    | 3                    |
| 7  | axon-modelling-5.0.3.jar                 | 7        | 92    | 2                    | 3                    |
| 8  | axon-eventsourcing-5.0.3.jar             | 7        | 100   | 3                    | 4                    |
| 9  | axon-common-5.0.3.jar                    | 15       | 156   | 10                   | 0                    |
| 10 | axon-messaging-5.0.3.jar                 | 59       | 579   | 7                    | 2                    |

## Table 1g - Top 30 artifacts with the lowest number of incoming dependencies

The following table lists the top 30 artifacts that are used the least by other artifacts (lowest count of incoming dependencies, lowest in-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-metrics-micrometer-5.0.3.jar        | 3        | 16    | 0                    | 3                    |
| 1  | axon-tracing-opentelemetry-5.0.3.jar     | 1        | 5     | 0                    | 2                    |
| 2  | axon-server-connector-5.0.3.jar          | 5        | 72    | 1                    | 4                    |
| 3  | axon-spring-boot-autoconfigure-5.0.3.jar | 7        | 75    | 1                    | 7                    |
| 4  | axon-test-5.0.3.jar                      | 6        | 78    | 1                    | 3                    |
| 5  | axon-update-5.0.3.jar                    | 5        | 23    | 1                    | 1                    |
| 6  | axon-modelling-5.0.3.jar                 | 7        | 92    | 2                    | 3                    |
| 7  | axon-eventsourcing-5.0.3.jar             | 7        | 100   | 3                    | 4                    |
| 8  | axon-conversion-5.0.3.jar                | 5        | 35    | 4                    | 1                    |
| 9  | axon-messaging-5.0.3.jar                 | 59       | 579   | 7                    | 2                    |
| 10 | axon-common-5.0.3.jar                    | 15       | 156   | 10                   | 0                    |

## Table 1h - Top 30 artifacts with the lowest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the lowest number of other artifacts (lowest count of outgoing dependencies, lowest out-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-common-5.0.3.jar                    | 15       | 156   | 10                   | 0                    |
| 1  | axon-conversion-5.0.3.jar                | 5        | 35    | 4                    | 1                    |
| 2  | axon-update-5.0.3.jar                    | 5        | 23    | 1                    | 1                    |
| 3  | axon-messaging-5.0.3.jar                 | 59       | 579   | 7                    | 2                    |
| 4  | axon-tracing-opentelemetry-5.0.3.jar     | 1        | 5     | 0                    | 2                    |
| 5  | axon-metrics-micrometer-5.0.3.jar        | 3        | 16    | 0                    | 3                    |
| 6  | axon-modelling-5.0.3.jar                 | 7        | 92    | 2                    | 3                    |
| 7  | axon-test-5.0.3.jar                      | 6        | 78    | 1                    | 3                    |
| 8  | axon-eventsourcing-5.0.3.jar             | 7        | 100   | 3                    | 4                    |
| 9  | axon-server-connector-5.0.3.jar          | 5        | 72    | 1                    | 4                    |
| 10 | axon-spring-boot-autoconfigure-5.0.3.jar | 7        | 75    | 1                    | 7                    |

## Cyclic Dependencies

Cyclic dependencies occur when one package uses a class of another package and vice versa. These dependencies can lead to problems when one of these packages needs to be changed.

## Table 2a - Cyclic Dependencies Overview

Show the top 40 cyclic dependencies sorted by the most promising to resolve first. This is done by calculating the number of forward dependencies (first cycle participant to second cycle participant) in relation to backward dependencies (second cycle participant back to first cycle participant). The higher this rate (approaching 1), the easier it should be to resolve the cycle by focussing on the few backward dependencies.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic\\_Dependencies](#)

### Columns:

- *artifactName* identifies the artifact of the first participant of the cycle
- *packageName* identifies the package of the first participant of the cycle
- *dependentArtifactName* identifies the artifact of the second participant of the cycle
- *dependentPackageName* identifies the package of the second participant of the cycle
- *forwardToBackwardBalance* is between 0 and 1. High for many forward and few backward dependencies.
- *numberForward* contains the number of dependencies from the first participant of the cycle to the second one
- *numberBackward* contains the number of dependencies from the second participant of the cycle back to the first one
- *someForwardDependencies* lists some forward dependencies in the text format "type1 -> type2"
- *backwardDependencies* lists the backward dependencies in the format "type1 <- type2" that are recommended to get resolved

|    | artifactName                | packageName                                                           | dependentArtifactName       | dependentPackageName                                    | forwardToBackwardBalance | numberForward | numberBack |
|----|-----------------------------|-----------------------------------------------------------------------|-----------------------------|---------------------------------------------------------|--------------------------|---------------|------------|
| 0  | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation                           | axon-messaging-5.0.3        | org.axonframework.messaging.core                        | 0.959184                 | 48            |            |
| 1  | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling                             | axon-messaging-5.0.3        | org.axonframework.messaging.core                        | 0.942857                 | 34            |            |
| 2  | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.annotation                  | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | 0.931034                 | 28            |            |
| 3  | axon-messaging-5.0.3        | org.axonframework.messaging.commandhandling.annotation                | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | 0.875000                 | 15            |            |
| 4  | axon-messaging-5.0.3        | org.axonframework.messaging.queryhandling.annotation                  | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | 0.857143                 | 13            |            |
| 5  | axon-modelling-5.0.3        | org.axonframework.modelling.annotation                                | axon-modelling-5.0.3        | org.axonframework.modelling                             | 0.846154                 | 12            |            |
| 6  | axon-messaging-5.0.3        | org.axonframework.messaging.core.unitofwork.transaction               | axon-messaging-5.0.3        | org.axonframework.messaging.core.unitofwork             | 0.666667                 | 5             |            |
| 7  | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | 0.666667                 | 15            |            |
| 8  | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | 0.666667                 | 15            |            |
| 9  | axon-common-5.0.3           | org.axonframework.common.configuration                                | axon-common-5.0.3           | org.axonframework.common.infra                          | 0.529412                 | 13            |            |
| 10 | axon-messaging-5.0.3        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | 0.500000                 | 15            |            |
| 11 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.event                          | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                  | 0.428571                 | 5             |            |
| 12 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                                | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.command          | 0.333333                 | 4             |            |
| 13 | axon-messaging-5.0.3        | org.axonframework.messaging.core.configuration                        | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | 0.333333                 | 2             |            |
| 14 | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling                             | axon-messaging-5.0.3        | org.axonframework.messaging.core.sequencing             | 0.333333                 | 4             |            |
| 15 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                  | 0.200000                 | 6             |            |
| 16 | axon-modelling-5.0.3        | org.axonframework.modelling.entity.annotation                         | axon-modelling-5.0.3        | org.axonframework.modelling.annotation                  | 0.200000                 | 3             |            |
| 17 | axon-modelling-5.0.3        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.3        | org.axonframework.modelling.entity                      | 0.142857                 | 8             |            |
| 18 | axon-messaging-5.0.3        | org.axonframework.messaging.core.unitofwork                           | axon-messaging-5.0.3        | org.axonframework.messaging.core                        | 0.111111                 | 10            |            |
| 19 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.util                           | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                  | 0.000000                 | 2             |            |

|    | artifactName             | packageName                                           | dependentArtifactName    | dependentPackageName                       | forwardToBackwardBalance | numberForward | numberBack |
|----|--------------------------|-------------------------------------------------------|--------------------------|--------------------------------------------|--------------------------|---------------|------------|
| 20 | axon-eventsourcing-5.0.3 | org.axonframework.eventsourcing.annotation.reflection | axon-eventsourcing-5.0.3 | org.axonframework.eventsourcing.annotation | 0.000000                 | 1             |            |

## Table 2b - Cyclic Dependencies Break Down

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic\_Dependencies\_Breakdown

### Columns in addition to Table 2a:

- dependency* shows the cycle dependency in the text format "type1 -> type2" (forward) or "type2<-type1" (backward)

[illegible]

Table 2c - Cyclic Dependencies Break Down - Backward Dependencies Only

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first. This table only contains the backward dependencies from the second participant of the cycle back to the first one that are the most promising to resolve.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic\_Dependencies\_Breakdown\_BackwardOnly

|    | artifactName                | packageName                                                           | dependentArtifactName       | dependentPackageName                                    | dependency forName                                                           |
|----|-----------------------------|-----------------------------------------------------------------------|-----------------------------|---------------------------------------------------------|------------------------------------------------------------------------------|
| 0  | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation                           | axon-messaging-5.0.3        | org.axonframework.messaging.core                        | HandlerAttributes<-SimpleHandlerAttributes                                   |
| 1  | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling                             | axon-messaging-5.0.3        | org.axonframework.messaging.core                        | EventMessage<-SubscribableEventSource                                        |
| 2  | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.annotation                  | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | EventHandler<-HandlerTypeResolver                                            |
| 3  | axon-messaging-5.0.3        | org.axonframework.messaging.commandhandling.annotation                | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | CommandHandler<-HandlerTypeResolver                                          |
| 4  | axon-messaging-5.0.3        | org.axonframework.messaging.queryhandling.annotation                  | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | QueryHandler<-HandlerTypeResolver                                            |
| 5  | axon-modelling-5.0.3        | org.axonframework.modelling.annotation                                | axon-modelling-5.0.3        | org.axonframework.modelling                             | TargetEntityIdMemberMismatchException<-PropertyBasedEntityIdResolver         |
| 6  | axon-messaging-5.0.3        | org.axonframework.messaging.core.unitofwork.transaction               | axon-messaging-5.0.3        | org.axonframework.messaging.core.unitofwork             | TransactionManager<-TransactionalUnitOfWorkFactory                           |
| 7  | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | PooledStreamingEventProcessorsConfigurer<-EventProcessingConfigurer          |
| 8  | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | PooledStreamingEventProcessorConfiguration<-EventProcessorModule             |
| 9  | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | PooledStreamingEventProcessorModule<-EventProcessorModule                    |
| 10 | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | SubscribingEventProcessorModule<-EventProcessorModule                        |
| 11 | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | SubscribingEventProcessorsConfigurer<-EventProcessingConfigurer              |
| 12 | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | SubscribingEventProcessorConfiguration<-EventProcessorModule                 |
| 13 | axon-common-5.0.3           | org.axonframework.common.configuration                                | axon-common-5.0.3           | org.axonframework.common.infra                          | Component<-JacksonComponentDescriptor                                        |
| 14 | axon-common-5.0.3           | org.axonframework.common.configuration                                | axon-common-5.0.3           | org.axonframework.common.infra                          | Component\$Identifier<-JacksonComponentDescriptor                            |
| 15 | axon-common-5.0.3           | org.axonframework.common.configuration                                | axon-common-5.0.3           | org.axonframework.common.infra                          | Component<-FileSystemStyleComponentDescriptor                                |
| 16 | axon-common-5.0.3           | org.axonframework.common.configuration                                | axon-common-5.0.3           | org.axonframework.common.infra                          | Component\$Identifier<-FileSystemStyleComponentDescriptor                    |
| 17 | axon-messaging-5.0.3        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | NoMoreInterceptors<-ChainedMessageHandlerInterceptorMember                   |
| 18 | axon-messaging-5.0.3        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | MessageHandlerInterceptorMemberChain<-AnnotatedHandlerInspector              |
| 19 | axon-messaging-5.0.3        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | NoMoreInterceptors<-AnnotatedHandlerInspector                                |
| 20 | axon-messaging-5.0.3        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | MessageInterceptingMember<-AnnotatedHandlerInspector                         |
| 21 | axon-messaging-5.0.3        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.3        | org.axonframework.messaging.core.annotation             | MessageHandlerInterceptorMemberChain<-ChainedMessageHandlerInterceptorMember |
| 22 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.event                          | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                  | EventProcessorControlService<-AxonServerConfigurationEnhancer                |
| 23 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.event                          | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                  | AxonServerEventStorageEngineFactory<-AxonServerConfigurationEnhancer         |
| 24 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                                | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.command          | MetadataConverter<-CommandConverter                                          |
| 25 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                                | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.command          | AxonServerConfiguration<-AxonServerCommandBusConnector                       |
| 26 | axon-messaging-5.0.3        | org.axonframework.messaging.core.configuration                        | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling.configuration | MessagingConfigurer<-EventProcessingConfigurer                               |
| 27 | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling                             | axon-messaging-5.0.3        | org.axonframework.messaging.core.sequencing             | EventMessage<-ExtractionSequencingPolicy                                     |
| 28 | axon-messaging-5.0.3        | org.axonframework.messaging.eventhandling                             | axon-messaging-5.0.3        | org.axonframework.messaging.core.sequencing             | EventMessage<-SequentialPerAggregatePolicy                                   |
| 29 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                  | AxonServerRemoteQueryHandlingException<-ErrorCode                            |
| 30 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                  | AxonServerNonTransientRemoteQueryHandlingException<-ErrorCode                |
| 31 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                  | AxonServerQueryDispatchException<-ErrorCode                                  |
| 32 | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.3 | org.axonframework.axonserver.connector                  | AxonServerQueryBusConnector<-AxonServerConfigurationEnhancer                 |
| 33 | axon-modelling-5.0.3        | org.axonframework.modelling.entity.annotation                         | axon-modelling-5.0.3        | org.axonframework.modelling.annotation                  | AnnotatedEntityMetamodel<-EntityIdResolverDefinition                         |
| 34 | axon-modelling-5.0.3        | org.axonframework.modelling.entity.annotation                         | axon-modelling-5.0.3        | org.axonframework.modelling.annotation                  | AnnotatedEntityMetamodel<-AnnotationBasedEntityIdResolverDefinition          |
| 35 | axon-modelling-5.0.3        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.3        | org.axonframework.modelling.entity                      | EntityChildMetamodel<-ConcreteEntityMetamodel                                |
| 36 | axon-modelling-5.0.3        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.3        | org.axonframework.modelling.entity                      | ChildAmbiguityException<-ConcreteEntityMetamodel                             |
| 37 | axon-modelling-5.0.3        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.3        | org.axonframework.modelling.entity                      | EntityChildMetamodel<-PolymorphicEntityMetamodel\$Builder                    |
| 38 | axon-modelling-5.0.3        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.3        | org.axonframework.modelling.entity                      | EntityChildMetamodel<-PolymorphicEntityMetamodel\$Builder                    |
| 39 | axon-modelling-5.0.3        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.3        | org.axonframework.modelling.entity                      | EntityChildMetamodel<-ConcreteEntityMetamodel\$Builder                       |



# Interface Segregation Candidates

Well known from [Design Principles and Design Patterns by Robert C. Martin](#), the *Interface Segregation Principle* suggests that software components should have narrow, focused interfaces rather than large, general-purpose ones. The goal is to minimize the dependencies between components and increase modularity, flexibility, and maintainability.

Smaller, focused and purpose-driven interfaces

- make it easier to modify individual components without affecting the rest of the system.
- make it clearer which client is affected by which change.
- don't force their clients to depend on methods they don't need.
- reduce the scope of changes since a change to one component doesn't affect others.
- lead to a more loosely coupled architecture that is easier to understand and maintain.

Reference: [Analyze java package metrics in a graph database](#)

## How to apply the results

If just one method of a type is used, especially in many places, then the result of this method can be used to call e.g. a method or construct an object instead of using the whole object and then just calling that single method.

If there are a couple of methods that are used for a distinct purpose, those could be factored out into a separate interface. The original type can extended/implement the new interface so that there are no breaking changes. Then all the callers, that use only this group of methods, can be changed to the new interface.

## Table 4 - Top 40 most used combinations of methods

The following table shows the top 40 most used combinations of methods of larger types that might benefit from applying the *Interface Segregation Principle*. The whole table can be found in the CSV report `Candidates_for_Interface_Segregation`.

|    | fullDependentTypeName                                                                                   | declaredMethods | calledMethodNames                                                                           | calledMethods | callerTypes |
|----|---------------------------------------------------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------|---------------|-------------|
| 0  | org.axonframework.messaging.core.unitofwork.ProcessingContext                                           | 32              | [computeResourceIfAbsent]                                                                   | 1             | 7           |
| 1  | org.axonframework.messaging.commandhandling.CommandBus                                                  | 6               | [dispatch]                                                                                  | 1             | 7           |
| 2  | org.axonframework.common.configuration.ComponentDefinition\$ComponentCreator                            | 17              | [createComponent]                                                                           | 1             | 5           |
| 3  | org.axonframework.messaging.core.MessageStream\$Entry                                                   | 7               | [message]                                                                                   | 1             | 5           |
| 4  | org.axonframework.messaging.core.unitofwork.ProcessingContext                                           | 32              | [withResource]                                                                              | 1             | 4           |
| 5  | org.axonframework.messaging.eventhandling.EventMessage                                                  | 19              | [identifier, timestamp]                                                                     | 2             | 4           |
| 6  | org.axonframework.messaging.eventhandling.EventMessage                                                  | 19              | [timestamp]                                                                                 | 1             | 4           |
| 7  | org.axonframework.messaging.core.MessageStream\$Empty                                                   | 44              | [cast]                                                                                      | 1             | 3           |
| 8  | org.axonframework.messaging.core.DelayedMessageStream                                                   | 42              | [create]                                                                                    | 1             | 3           |
| 9  | org.axonframework.messaging.eventhandling.EventMessage                                                  | 19              | [andMetadata]                                                                               | 1             | 3           |
| 10 | org.axonframework.messaging.core.annotation.WrappedMessageHandlingMember                                | 13              | [handleSync]                                                                                | 1             | 3           |
| 11 | org.axonframework.messaging.core.unitofwork.UnitOfWork                                                  | 24              | [execute]                                                                                   | 1             | 2           |
| 12 | org.axonframework.test.fixture.AxonTestThenMessage                                                      | 22              | [exceptionSatisfies, exception]                                                             | 3             | 2           |
| 13 | org.axonframework.messaging.commandhandling.CommandMessage                                              | 20              | [routingKey, priority]                                                                      | 2             | 2           |
| 14 | org.axonframework.messaging.commandhandling.CommandMessage                                              | 19              | [withConvertedPayload]                                                                      | 1             | 2           |
| 15 | org.axonframework.messaging.commandhandling.CommandMessage                                              | 19              | [andMetadata]                                                                               | 1             | 2           |
| 16 | org.axonframework.messaging.eventhandling.EventMessage                                                  | 19              | [withConvertedPayload]                                                                      | 1             | 2           |
| 17 | org.axonframework.messaging.eventhandling.EventMessage                                                  | 19              | [identifier]                                                                                | 1             | 2           |
| 18 | org.axonframework.messaging.eventstreaming.EventCriteria                                                | 12              | [tags]                                                                                      | 1             | 2           |
| 19 | org.axonframework.messaging.eventstreaming.OrEventCriteria                                              | 12              | [or]                                                                                        | 1             | 2           |
| 20 | org.axonframework.modelling.entity.ConcreteEntityMetamodel                                              | 11              | [forEntityClass]                                                                            | 1             | 2           |
| 21 | org.axonframework.modelling.entity.PolymorphicEntityMetamodel                                           | 11              | [forSuperType]                                                                              | 1             | 2           |
| 22 | org.axonframework.event sourcing.eventstore.EventStore                                                  | 9               | [transaction]                                                                               | 1             | 2           |
| 23 | org.axonframework.messaging.eventhandling.processing.streaming.token.WrappedToken                       | 9               | [unwrapLowerBound]                                                                          | 1             | 2           |
| 24 | org.axonframework.messaging.eventhandling.processing.streaming.token.store.ConfigToken                  | 9               | [get]                                                                                       | 1             | 2           |
| 25 | org.axonframework.messaging.core.MessageStream\$Entry                                                   | 7               | [map]                                                                                       | 1             | 2           |
| 26 | org.axonframework.conversion.ChainingContentTypeConverter                                               | 5               | [canConvert, registerConverter]                                                             | 2             | 2           |
| 27 | org.axonframework.messaging.core.EmptyMessageStream                                                     | 44              | [instance]                                                                                  | 1             | 1           |
| 28 | org.axonframework.event sourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream | 43              | [isCompleted, hasNextAvailable]                                                             | 2             | 1           |
| 29 | org.axonframework.event sourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream | 42              | [callback]                                                                                  | 1             | 1           |
| 30 | org.axonframework.messaging.core.DelayedMessageStream                                                   | 42              | [createSingle]                                                                              | 1             | 1           |
| 31 | org.axonframework.messaging.core.MessageStream\$Single                                                  | 42              | [asCompletableFuture]                                                                       | 1             | 1           |
| 32 | org.axonframework.messaging.core.MessageStream\$Single                                                  | 42              | [first]                                                                                     | 1             | 1           |
| 33 | org.axonframework.messaging.core.MessageStream\$Single                                                  | 42              | [cast]                                                                                      | 1             | 1           |
| 34 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                           | 37              | [removeResource, putResourceIfAbsent, updateResource, putResource, computeResourceIfAbsent] | 6             | 1           |
| 35 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                           | 36              | [updateResource, computeResourceIfAbsent]                                                   | 2             | 1           |
| 36 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                           | 35              | [computeResourceIfAbsent]                                                                   | 1             | 1           |
| 37 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                           | 34              | [computeResourceIfAbsent]                                                                   | 1             | 1           |
| 38 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                           | 33              | [putResource]                                                                               | 1             | 1           |
| 39 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                           | 33              | [removeResource, computeResourceIfAbsent]                                                   | 2             | 1           |

## Package Usage

Table 5 - Types that are used by multiple packages

This table shows the top 40 packages that are used by the highest number of different packages. The whole table can be found in the CSV report `List_types_that_are_used_by_many_different_packages` .

|    | fullQualifiedDependentTypeName                                       | dependentTypeName        | dependentTypeLabels                                                                                                                                                                                                                                                                                                                                                                                                                                            | numberOfUsingPackages |
|----|----------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 0  | org.axonframework.messaging.core.unitofwork.ProcessingContext        | ProcessingContext        | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...                                                                                                                                                    | 59                    |
| 1  | org.axonframework.common.annotation.Internal                         | Internal                 | [Type, File, Java, ByteCode, Annotation, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent...                                                                                                                     | 55                    |
| 2  | org.axonframework.messaging.core.Message                             | Message                  | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...                                                                                                                                                    | 47                    |
| 3  | org.axonframework.common.infra.ComponentDescriptor                   | ComponentDescriptor      | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...                                                                                                                     | 46                    |
| 4  | org.axonframework.messaging.core.MessageStream                       | MessageStream            | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWea...                                                                                                                       | 36                    |
| 5  | org.axonframework.messaging.eventhandling.EventMessage               | EventMessage             | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponen...                                                                                                                       | 35                    |
| 6  | org.axonframework.messaging.core.MessageType                         | MessageType              | [Type, File, Java, ByteCode, Record, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...                                                                                                                                                    | 31                    |
| 7  | org.axonframework.messaging.core.QualifiedName                       | QualifiedName            | [Type, File, Java, ByteCode, Record, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4T...                                                                                                                                                    | 30                    |
| 8  | org.axonframework.common.configuration.Configuration                 | Configuration            | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity4,...                                                            | 29                    |
| 9  | org.axonframework.common.FutureUtils                                 | FutureUtils              | [Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity6, Mark4...                                                          | 28                    |
| 10 | org.axonframework.messaging.core.Context\$ResourceKey                | Context\$ResourceKey     | [Type, File, Java, ByteCode, GenericDeclaration, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyCon...                                                                                                                     | 25                    |
| 11 | org.axonframework.messaging.core.MessageStream\$Empty                | MessageStream\$Empty     | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation12, Mark4TypeLouvainCommunity9, Mark4TypeLeidenCommunity10, Mark4Type... | 25                    |
| 12 | org.axonframework.common.infra.DescribableComponent                  | DescribableComponent     | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...                                                                                                                     | 24                    |
| 13 | org.axonframework.common.Assert                                      | Assert                   | [Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation5, Mark4TypeLouvainCommunity11, Mark4TypeLeidenCommunity3, Mark4TypeCoreDecompositi...                                                                                                                     | 22                    |
| 14 | org.axonframework.common.configuration.ComponentRegistry             | ComponentRegistry        | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut1, Mark4TypeHBSCAN63]                                                                                                                               | 22                    |
| 15 | org.axonframework.common.BuilderUtils                                | BuilderUtils             | [Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation22, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity8, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut33, Mark4TypeHBSCAN41, Mark4TopAnomalyHub]                                                                                                              | 21                    |
| 16 | org.axonframework.messaging.commandhandling.CommandMessage           | CommandMessage           | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity2, ...                                | 20                    |
| 17 | org.axonframework.messaging.core.annotation.ParameterResolverFactory | ParameterResolverFactory | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity13, Mark4TypeLeidenCommunity9, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut53, Mark4TypeHBSCAN97]                                                                                                                              | 20                    |
| 18 | org.axonframework.messaging.core.MessageStream\$Entry                | MessageStream\$Entry     | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0,                                                                                                   | 19                    |

|    | fullQualifiedDependentTypeName                                                            | dependentTypeName                                  | dependentTypeLabels                                                                                                                                                                                                                                                                                         | numberOfUsingPackages |
|----|-------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |                                                                                           |                                                    | Mark4TypeLabelPropagation12, Mark4TypeLouvainCommunity9, Mark4T...                                                                                                                                                                                                                                          |                       |
| 19 | org.axonframework.common.configuration.ConfigurationEnhancer                              | ConfigurationEnhancer                              | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut98, Mark4TypeHDBSCAN133]                                                                    | 18                    |
| 20 | org.axonframework.conversion.Converter                                                    | Converter                                          | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecompo...  | 18                    |
| 21 | org.axonframework.messaging.core.MessageStream\$Single                                    | MessageStream\$Single                              | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation12, Mar...                                                   | 18                    |
| 22 | org.axonframework.messaging.core.MessageTypeResolver                                      | MessageTypeResolver                                | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition10, ...  | 18                    |
| 23 | org.axonframework.messaging.core.conversion.MessageConverter                              | MessageConverter                                   | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut9, Mark4TypeLocalClusteringCoefficient0.11333333333333333, Mark4TypeHDBSCAN111]             | 17                    |
| 24 | org.axonframework.messaging.core.Metadata                                                 | Metadata                                           | [Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation0, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut83, Ma...  | 16                    |
| 25 | org.axonframework.common.ObjectUtils                                                      | ObjectUtils                                        | [Type, File, Java, ByteCode, Class, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut64, Mark4TypeHDBSCAN-1]                                                                         | 16                    |
| 26 | org.axonframework.messaging.eventhandling.processing.streaming.token.TrackingToken        | TrackingToken                                      | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation15, Mark4TypeLouvainCommunity6... | 15                    |
| 27 | org.axonframework.common.AxonConfigurationException                                       | AxonConfigurationException                         | [Type, File, Java, ByteCode, Class, Throwable, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation30, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN54]                               | 14                    |
| 28 | org.axonframework.messaging.commandhandling.CommandBus                                    | CommandBus                                         | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity2, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut15, Mark4TypeHDBSCAN71]                                                                    | 13                    |
| 29 | org.axonframework.common.ReflectionUtils                                                  | ReflectionUtils                                    | [Type, File, Java, ByteCode, Class, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation5, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity3, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut90, Mark4TypeHDBSCAN49, Mark4TopAnomalyBottleneck, Mark4TopA...  | 13                    |
| 30 | org.axonframework.messaging.commandhandling.CommandResultMessage                          | CommandResultMessage                               | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity2, Mark4TypeCoreDecomposition10, ...  | 12                    |
| 31 | org.axonframework.common.configuration.ComponentDefinition                                | ComponentDefinition                                | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut24, Mark4TypeHDBSCAN-1]                                               | 12                    |
| 32 | org.axonframework.common.configuration.ComponentDefinition\$IncompleteComponentDefinition | ComponentDefinition\$IncompleteComponentDefinition | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut62, Mark4TypeHDBSCAN-1]                                               | 12                    |
| 33 | org.axonframework.messaging.core.Context                                                  | Context                                            | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation12, Mark4TypeLouvainCommunity9, Mark4TypeLeidenC...                                | 12                    |
| 34 | org.axonframework.messaging.queryhandling.QueryMessage                                    | QueryMessage                                       | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation62, Mark4TypeLouvainCommunity9, Mark4TypeLeidenCommunity10, Mark4TypeCoreDecomposition10...  | 12                    |
| 35 | org.axonframework.common.AxonNonTransientException                                        | AxonNonTransientException                          | [Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation5, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity7, Mark4TypeCoreDecomposition4, Mark4TypeMaximumKCut20, Mark4TypeLocalClusteringCoef...  | 11                    |
| 36 | org.axonframework.common.configuration.ComponentBuilder                                   | ComponentBuilder                                   | [Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity6,                                                                                                                          | 11                    |

|    | fullQualifiedDependentTypeName                                      | dependentTypeName         | dependentTypeLabels                                                                                                                                                                                                                                                                                                           | numberOfUsingPackages |
|----|---------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |                                                                     |                           | Mark4TypeKCoreDecomposition10,<br>Mark4TypeMaximumKCut35, Mark4TypeHDBSCAN125]                                                                                                                                                                                                                                                |                       |
| 37 | org.axonframework.messaging.eventhandling.conversion.EventConverter | EventConverter            | [Type, File, Java, ByteCode, Interface,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation9,<br>Mark4TypeLouvainCommunity13,<br>Mark4TypeLeidenCommunity2,<br>Mark4TypeKCoreDecomposition9,<br>Mark4TypeMaximumKCut29, Mark4TypeHDBSCAN147]                                                                  | 11                    |
| 38 | org.axonframework.messaging.core.MessageHandlerInterceptor          | MessageHandlerInterceptor | [Type, File, Java, ByteCode, Interface, GenericDeclaration,<br>Mark4TopCentralityHyperlinkInducedTopicSearchAuthority,<br>Mark4TopCentralityHyperlinkInducedTopicSearchHub,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation9,<br>Mark4TypeLouvainCommunity4,<br>Mark4TypeLeidenCommunity6, Mark4TypeKC... | 11                    |
| 39 | org.axonframework.messaging.core.annotation.ParameterResolver       | ParameterResolver         | [Type, File, Java, ByteCode, Interface, GenericDeclaration,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation7,<br>Mark4TypeLouvainCommunity3,<br>Mark4TypeLeidenCommunity4,<br>Mark4TypeKCoreDecomposition7,<br>Mark4TypeMaximumKCut61, Mark4TypeHDBSCAN128]                                               | 11                    |

Table 6 - Packages that are used by multiple artifacts

This table shows the top 30 artifacts that only use a few (compared to all existing) packages of another artifact. The whole table can be found in the CSV report `ArtifactPackageUsage`.

|    | artifactName                         | dependentArtifactName                | dependentPackages | dependentArtifactPackages | packageUsagePercentage | dependentFullQualifiedPackageNames                                                                                                                                                                                                                                                                             | dependentPackageNames                                                                                                                                                      |
|----|--------------------------------------|--------------------------------------|-------------------|---------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0  | axon-tracing-opentelemetry-5.0.3     | axon-messaging-5.0.3                 | 2                 | 59                        | 0.033898               | [org.axonframework.messaging.core, org.axonframework.messaging.tracing]                                                                                                                                                                                                                                        | [core, tracing]                                                                                                                                                            |
| 1  | axon-tracing-opentelemetry-5.0.3     | axon-common-5.0.3                    | 1                 | 15                        | 0.066667               | [org.axonframework.common]                                                                                                                                                                                                                                                                                     | [common]                                                                                                                                                                   |
| 2  | axon-metrics-micrometer-5.0.3        | axon-messaging-5.0.3                 | 7                 | 59                        | 0.118644               | [org.axonframework.messaging.core, org.axonframework.messaging.eventhandling.processing, org.axonframework.messaging.monitoring, org.axonframework.messaging.commandhandling, org.axonframework.messaging.queryhandling, org.axonframework.messaging.eventhandling, org.axonframework.messaging.monitori...    | [core, processing, monitoring, commandhandling, queryhandling, eventhandling, configuration]                                                                               |
| 3  | axon-metrics-micrometer-5.0.3        | axon-common-5.0.3                    | 2                 | 15                        | 0.133333               | [org.axonframework.common.configuration, org.axonframework.common]                                                                                                                                                                                                                                             | [configuration, common]                                                                                                                                                    |
| 4  | axon-test-5.0.3                      | axon-eventsourcing-5.0.3             | 1                 | 7                         | 0.142857               | [org.axonframework.eventsourcing.eventstore]                                                                                                                                                                                                                                                                   | [eventstore]                                                                                                                                                               |
| 5  | axon-server-connector-5.0.3          | axon-eventsourcing-5.0.3             | 1                 | 7                         | 0.142857               | [org.axonframework.eventsourcing.eventstore]                                                                                                                                                                                                                                                                   | [eventstore]                                                                                                                                                               |
| 6  | axon-server-connector-5.0.3          | axon-modelling-5.0.3                 | 1                 | 7                         | 0.142857               | [org.axonframework.modelling]                                                                                                                                                                                                                                                                                  | [modelling]                                                                                                                                                                |
| 7  | axon-metrics-micrometer-5.0.3        | axon-spring-boot-autoconfigure-5.0.3 | 1                 | 7                         | 0.142857               | [org.axonframework.extension.springboot.autoconfig]                                                                                                                                                                                                                                                            | [autoconfig]                                                                                                                                                               |
| 8  | axon-test-5.0.3                      | axon-messaging-5.0.3                 | 9                 | 59                        | 0.152542               | [org.axonframework.messaging.commandhandling, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core.conversion, org.axonframework.messaging.eventstreaming, org.axonframework.messaging.eventhandling.processing.streaming.token, org.axonframew...    | [commandhandling, core, eventstreaming, conversion, token, annotation, monitoring]                                                                                         |
| 9  | axon-spring-boot-autoconfigure-5.0.3 | axon-test-5.0.3                      | 1                 | 6                         | 0.166667               | [org.axonframework.test.server]                                                                                                                                                                                                                                                                                | [server]                                                                                                                                                                   |
| 10 | axon-modelling-5.0.3                 | axon-conversion-5.0.3                | 1                 | 5                         | 0.200000               | [org.axonframework.conversion]                                                                                                                                                                                                                                                                                 | [conversion]                                                                                                                                                               |
| 11 | axon-conversion-5.0.3                | axon-common-5.0.3                    | 3                 | 15                        | 0.200000               | [org.axonframework.common.infra, org.axonframework.common.annotation, org.axonframework.common]                                                                                                                                                                                                                | [infra, annotation, common]                                                                                                                                                |
| 12 | axon-eventsourcing-5.0.3             | axon-conversion-5.0.3                | 1                 | 5                         | 0.200000               | [org.axonframework.conversion]                                                                                                                                                                                                                                                                                 | [conversion]                                                                                                                                                               |
| 13 | axon-conversion-5.0.3                | axon-conversion-5.0.3                | 1                 | 5                         | 0.200000               | [org.axonframework.conversion]                                                                                                                                                                                                                                                                                 | [conversion]                                                                                                                                                               |
| 14 | axon-update-5.0.3                    | axon-common-5.0.3                    | 3                 | 15                        | 0.200000               | [org.axonframework.common.annotation, org.axonframework.common.configuration]                                                                                                                                                                                                                                  | [annotation, common, configuration]                                                                                                                                        |
| 15 | axon-spring-boot-autoconfigure-5.0.3 | axon-server-connector-5.0.3          | 1                 | 5                         | 0.200000               | [org.axonframework.axonserver.connector]                                                                                                                                                                                                                                                                       | [connector]                                                                                                                                                                |
| 16 | axon-modelling-5.0.3                 | axon-messaging-5.0.3                 | 13                | 59                        | 0.220339               | [org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core, org.axonframework.messaging.commandhandling, org.axonframework.messaging.queryhandling.configuration, org.axonframework.messaging.commandhandling.configuration, org.axonfram...    | [unitofwork, eventhandling, core, commandhandling, configuration, conversion, annotation, reflection]                                                                      |
| 17 | axon-test-5.0.3                      | axon-common-5.0.3                    | 4                 | 15                        | 0.266667               | [org.axonframework.common, org.axonframework.common.configuration, org.axonframework.common.infra, org.axonframework.common.annotation]                                                                                                                                                                        | [common, configuration, infra, annotation]                                                                                                                                 |
| 18 | axon-server-connector-5.0.3          | axon-messaging-5.0.3                 | 16                | 59                        | 0.271186               | [org.axonframework.messaging.eventhandling.processing.subscribing, org.axonframework.messaging.eventstreaming, org.axonframework.messaging.eventhandling.processing.streaming, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling.processing, org.axonframework.messaging.event...    | [subscribing, eventstreaming, streaming, core, processing, token, conversion, unitofwork, segmentation, store, eventhandling, distributed, commandhandling, queryhandling] |
| 19 | axon-eventsourcing-5.0.3             | axon-messaging-5.0.3                 | 16                | 59                        | 0.271186               | [org.axonframework.messaging.eventhandling.annotation, org.axonframework.messaging.eventstreaming, org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling.processing.streaming.token, org.a...    | [annotation, eventstreaming, unitofwork, eventhandling, core, token, conversion, commandhandling, configuration, interception, transaction]                                |
| 20 | axon-spring-boot-autoconfigure-5.0.3 | axon-messaging-5.0.3                 | 16                | 59                        | 0.271186               | [org.axonframework.messaging.core.timeout, org.axonframework.messaging.core.conversion, org.axonframework.messaging.core.unitofwork.transaction.jpaa, org.axonframework.messaging.core.annotation, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling, org.axonframework.messaging... | [timeout, conversion, jpaa, annotation, core, eventhandling, correlation, interception, commandhandling, distributed, queryhandling, store, transaction]                   |
| 21 | axon-spring-boot-autoconfigure-5.0.3 | axon-eventsourcing-5.0.3             | 2                 | 7                         | 0.285714               | [org.axonframework.eventsourcing.eventstore.jpaa, org.axonframework.eventsourcing.eventstore]                                                                                                                                                                                                                  | [jpaa, eventstore]                                                                                                                                                         |
| 22 | axon-spring-boot-autoconfigure-5.0.3 | axon-common-5.0.3                    | 5                 | 15                        | 0.333333               | [org.axonframework.common.annotation, org.axonframework.common.jpaa, org.axonframework.common.configuration, org.axonframework.common]                                                                                                                                                                         | [annotation, jpaa, jdbc, configuration, common]                                                                                                                            |
| 23 | axon-modelling-5.0.3                 | axon-common-5.0.3                    | 5                 | 15                        | 0.333333               | [org.axonframework.common.infra, org.axonframework.common.configuration, org.axonframework.common.property, org.axonframework.common.annotation]                                                                                                                                                               | [infra, common, configuration, property, annotation]                                                                                                                       |
| 24 | axon-server-connector-5.0.3          | axon-common-5.0.3                    | 6                 | 15                        | 0.400000               | [org.axonframework.common, org.axonframework.common.annotation, org.axonframework.common.configuration, org.axonframework.common.lifecycle, org.axonframework.common.util]                                                                                                                                     | [common, annotation, configuration, infra, lifecycle, util]                                                                                                                |
| 25 | axon-messaging-5.0.3                 | axon-conversion-5.0.3                | 2                 | 5                         | 0.400000               | [org.axonframework.conversion, org.axonframework.conversion.jackson]                                                                                                                                                                                                                                           | [conversion, jackson]                                                                                                                                                      |
| 26 | axon-common-5.0.3                    | axon-common-5.0.3                    | 8                 | 15                        | 0.533333               | [org.axonframework.common, org.axonframework.common.tx, org.axonframework.common.function, org.axonframework.common.annotation, org.axonframework.common.configuration, org.axonframework.common.lifecycle, org.axonframework.common.io]                                                                       | [common, tx, function, annotation, configuration, infra, lifecycle, io]                                                                                                    |
| 27 | axon-eventsourcing-5.0.3             | axon-common-5.0.3                    | 8                 | 15                        | 0.533333               | [org.axonframework.common.infra, org.axonframework.common.configuration, org.axonframework.common, org.axonframework.common.jpaa, org.axonframework.common.tx, org.axonframework.common.jdbc, org.axonframework.common.io]                                                                                     | [infra, configuration, common, annotation, jpaa, tx, jdbc, io]                                                                                                             |
| 28 | axon-spring-boot-autoconfigure-5.0.3 | axon-update-5.0.3                    | 3                 | 5                         | 0.600000               | [org.axonframework.update.configuration, org.axonframework.update.detection, org.axonframework.update]                                                                                                                                                                                                         | [configuration, detection, update]                                                                                                                                         |
| 29 | axon-test-5.0.3                      | axon-test-5.0.3                      | 4                 | 6                         | 0.666667               | [org.axonframework.test, org.axonframework.test.util, org.axonframework.test.matchers, org.axonframework.test.fixture]                                                                                                                                                                                         | [test, util, matchers, fixture]                                                                                                                                            |

Table 7 - Types that are used by multiple artifacts

This table shows the top 30 types that only use a few (compared to all existing) types of another artifact. The whole table can be found in the CSV report `ClassesPerPackageUsageAcrossArtifacts`.

|    | artifactName                     | dependentArtifactName                | packageName                                                                | dependentPackage.fqn                              | dependentTypes | dependentPackageTypes | typeUsagePercentage |
|----|----------------------------------|--------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------|----------------|-----------------------|---------------------|
| 0  | axon-modelling-5.0.3             | axon-messaging-5.0.3                 | org.axonframework.modelling.repository                                     | org.axonframework.messaging.core                  | 1              | 80                    | 0.0%                |
| 1  | axon-test-5.0.3                  | axon-messaging-5.0.3                 | org.axonframework.test.matchers                                            | org.axonframework.messaging.core                  | 1              | 80                    | 0.0%                |
| 2  | axonsourcing-5.0.3               | axon-messaging-5.0.3                 | org.axonframework.eventsourcing.configuration                              | org.axonframework.messaging.core                  | 1              | 80                    | 0.0%                |
| 3  | axonsourcing-5.0.3               | axon-messaging-5.0.3                 | org.axonframework.eventsourcing.configuration                              | org.axonframework.messaging.core.annotation       | 1              | 51                    | 0.0%                |
| 4  | axonsourcing-5.0.3               | axon-common-5.0.3                    | org.axonframework.eventsourcing.annotation                                 | org.axonframework.common.configuration            | 1              | 46                    | 0.0%                |
| 5  | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.core.unitofwork                                | org.axonframework.common.configuration            | 1              | 46                    | 0.0%                |
| 6  | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.core                                           | org.axonframework.common.configuration            | 1              | 46                    | 0.0%                |
| 7  | axon-modelling-5.0.3             | axon-common-5.0.3                    | org.axonframework.modelling.entity.annotation                              | org.axonframework.common.configuration            | 1              | 46                    | 0.0%                |
| 8  | axonsourcing-5.0.3               | axon-common-5.0.3                    | org.axonframework.eventsourcing.annotation.reflection                      | org.axonframework.common.configuration            | 1              | 46                    | 0.0%                |
| 9  | axon-metrics-micrometer-5.0.3    | axon-spring-boot-autoconfigure-5.0.3 | org.axonframework.extension.metrics.micrometer.springboot                  | org.axonframework.extension.springboot.autoconfig | 1              | 41                    | 0.0%                |
| 10 | axon-tracing-opentelemetry-5.0.3 | axon-messaging-5.0.3                 | org.axonframework.extension.tracing.opentelemetry                          | org.axonframework.messaging.core                  | 2              | 80                    | 0.0%                |
| 11 | axon-conversion-5.0.3            | axon-common-5.0.3                    | org.axonframework.conversion.avro                                          | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 12 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.commandhandling.tracing                        | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 13 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.eventhandling.processing.streaming.token.store | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 14 | axon-test-5.0.3                  | axon-common-5.0.3                    | org.axonframework.test.util                                                | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 15 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.core.interception.annotation                   | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 16 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.core.unitofwork.annotation                     | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 17 | axon-modelling-5.0.3             | axon-common-5.0.3                    | org.axonframework.modelling.repository                                     | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 18 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.commandhandling.interception                   | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 19 | axon-tracing-opentelemetry-5.0.3 | axon-common-5.0.3                    | org.axonframework.extension.tracing.opentelemetry                          | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 20 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.monitoring.configuration                       | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 21 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.eventstreaming                                 | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 22 | axon-test-5.0.3                  | axon-common-5.0.3                    | org.axonframework.test.server                                              | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 23 | axon-test-5.0.3                  | axon-common-5.0.3                    | org.axonframework.test.extension                                           | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 24 | axon-update-5.0.3                | axon-common-5.0.3                    | org.axonframework.update                                                   | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 25 | axonsourcing-5.0.3               | axon-common-5.0.3                    | org.axonframework.eventsourcing.annotation                                 | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 26 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.eventhandling.processing                       | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 27 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.eventhandling.tracing                          | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 28 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.monitoring                                     | org.axonframework.common                          | 1              | 34                    | 0.0%                |
| 29 | axon-messaging-5.0.3             | axon-common-5.0.3                    | org.axonframework.messaging.queryhandling.configuration                    | org.axonframework.common                          | 1              | 34                    | 0.0%                |

Table 8 - Duplicate package names across artifacts

This table shows the top 30 duplicate package names across artifacts. They are ordered by the number of duplicates descending.

This might lead to confusion, makes importing more error prone and might even lead to duplicate classes where only one of them will be loaded by the class loader. If a package is named the same way in two or more artifacts this even allows another artifact to access package protected classes, methods or members which might not be intended.

The whole table can be found in the CSV report `DuplicatePackageNamesAcrossArtifacts`.

| packageName | duplicates | artifactNames |
|-------------|------------|---------------|
|-------------|------------|---------------|

Table 9 - Annotated elements

This table shows 30 most used Java Annotations including some examples where they are used.





| dependency.fileDistanceAsFewestChangeDirectoryCommands | numberOfDependencies | numberOfDependencyUsers | numberOfDependencyProviders |     |
|--------------------------------------------------------|----------------------|-------------------------|-----------------------------|-----|
| 0                                                      | 0                    | 2125                    | 881                         | 921 |
| 1                                                      | 1                    | 100                     | 86                          | 41  |
| 2                                                      | 2                    | 2138                    | 631                         | 420 |
| 3                                                      | 4                    | 2104                    | 679                         | 322 |